

Exercises 330–336 and Problems 17–23 Detailed Solutions

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Exercise 330

Claim. If A is Hermitian positive semi-definite, then its diagonal entries are non-negative.

Proof. For any index i , choose the standard basis vector e_i . Since A is PSD,

$$e_i^H A e_i \geq 0.$$

But $e_i^H A e_i = a_{ii}$, so $a_{ii} \geq 0$ for all i . □

Exercise 331

Claim. Every principal submatrix of a PSD matrix is PSD.

Proof. Let B be a principal submatrix of A . Then there exists a permutation matrix P such that

$$A = P \begin{pmatrix} B & * \\ * & * \end{pmatrix} P^T.$$

For any y , define $z = P^T(y, 0)^T$. Then

$$z^H A z = (y, 0)^H \begin{pmatrix} B & * \\ * & * \end{pmatrix} (y, 0) = y^H B y \geq 0.$$

Thus B is PSD. □

Exercise 332

Claim. A Hermitian PSD matrix has non-negative eigenvalues.

Proof. Let $Av = \lambda v$. Then

$$v^H A v = \lambda v^H v.$$

Since A is PSD, $v^H A v \geq 0$ and $v^H v > 0$, so $\lambda \geq 0$. □

Exercise 333

Claim. AA^H is Hermitian PSD for any matrix A .

Proof. Clearly $(AA^H)^H = AA^H$. For PSD,

$$x^H AA^H x = \|A^H x\|^2 \geq 0.$$

Thus AA^H is Hermitian PSD. □

Exercise 334

Claim. Every Hermitian PSD matrix H can be written as AA^H .

Proof. Spectral theorem gives $H = U\Lambda U^H$, where Λ is diagonal with non-negative entries. Let $\Lambda^{1/2}$ be square roots. Set $A = U\Lambda^{1/2}$. Then

$$AA^H = U\Lambda^{1/2}(U\Lambda^{1/2})^H = H. \square$$

Exercise 335

For Hermitian positive definite matrices:

- Diagonal entries > 0 .
- All principal submatrices are positive definite.
- All eigenvalues are strictly positive.
- Factorization $A = GG^H$ exists with G nonsingular.

Exercise 336 — Cholesky Factorization

Let

$$A = \begin{pmatrix} \alpha & a^H \\ a & B \end{pmatrix},$$

where $\alpha > 0$ and $B - \frac{1}{\alpha}aa^H$ is HPD. By induction, there exists $\tilde{G}$ such that

$$B - \frac{1}{\alpha}aa^H = \tilde{G}\tilde{G}^H.$$

Define

$$G = \begin{pmatrix} \sqrt{\alpha} & 0 \\ \frac{1}{\sqrt{\alpha}}a & \tilde{G} \end{pmatrix}.$$

Then $A = GG^H$. □

Problem 17

Let

$$B = \begin{pmatrix} 0 & A^T \\ A & 0 \end{pmatrix}.$$

Then $B^T = B$, so B is symmetric. If $A = U\Sigma V^T$ (SVD), define

$$Q = \begin{pmatrix} V & 0 \\ 0 & U \end{pmatrix}.$$

Then

$$Q^T B Q = \begin{pmatrix} 0 & \Sigma^T \\ \Sigma & 0 \end{pmatrix},$$

which becomes block diagonal (2-by-2 blocks) under permutation. Thus the Schur decomposition of B is derived from the SVD of A .

Problem 18

Show $|\sigma_i(A + E) - \sigma_i(A)| \leq \|E\|_2$.

Proof. Using min-max characterization of singular values,

$$\sigma_i(A + E) \leq \sigma_i(A) + \|E\|_2, \quad \sigma_i(A) \leq \sigma_i(A + E) + \|E\|_2.$$

Thus

$$|\sigma_i(A + E) - \sigma_i(A)| \leq \|E\|_2. \square$$

Problem 19

The best rank- k approximation satisfies

$$\min_{\text{rank}(B) \leq k} \|A - B\|_2 = \sigma_{k+1}(A).$$

By SVD $A_k = \sum_{i=1}^k \sigma_i u_i v_i^T$ achieves the minimum.

Problem 20

If

$$A = \begin{pmatrix} B \\ c^T \end{pmatrix},$$

then

$$A^T A = B^T B + cc^T,$$

a rank-one PSD update. Hence eigenvalues interlace, giving

$$\cdots \leq \sigma_k(B) \leq \sigma_k(A) \leq \sigma_{k-1}(B) \leq \cdots \leq \sigma_1(A).$$

Problem 21

If $A = U\Sigma V^H$ is complex SVD, then

$$T(A) = T(U) \begin{pmatrix} \Sigma & 0 \\ 0 & \Sigma \end{pmatrix} T(V)^T$$

is the real SVD, since $T(U), T(V)$ are orthogonal.

Problem 22 — Wielandt–Hoffman

For normal matrices,

$$\min_{\sigma} \sum_{i=1}^n |\lambda_i(A) - \lambda_{\sigma(i)}(B)|^2 \leq \|A - B\|_F^2.$$

This follows from unitary invariance and majorization in diagonal case.

Problem 23

$$\min_X \|AX - I\|_F = \|AA^+ - I\|_F.$$

Proof. Write $AX - I = [Ax_1 - e_1, \dots, Ax_m - e_m]$. Each column is an independent least-squares problem, minimized by $x_j = A^+ e_j$. Hence the minimizer is $X = A^+$ and

$$\|AX - I\|_F^2 = \|AA^+ - I\|_F^2.$$

Taking square roots completes the proof. □